Syllabic frequencies reveal a strong cross-lingusitic prevalence of the Sonority Sequencing Principle

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Sonority is a phonological construct that describes the perceptual salience of a given speech sound relative to neighbouring ones (Blevins, 1995; S. Parker, 2008; S. G. Parker, 2002; Selkirk, 1984). It is conceived as a universal hierarchy of phoneme categories such that categories in the lower and higher end of the hierarchy bear lower and higher sonority, respectively (e.g., *Obstruents* < *Nasals* < *Liquids* < *Glides* < *Vowels*, Clements, 1990). Sonority has proven useful as a framework to explain why, across many different languages, some syllables are considered well-formed while others are marked as ill-formed. This effort has derived into many proposed different sonority-based rules. One notable example is the Sonority Sequencing Principle (SSP), a phonotactical rule that constrains the ordering of speech sounds that form a syllable based on sonority. The SSP states that each syllable displays a unique peak of sonority in the nucleus, and that sonority rises monotonically from the syllable margins toward the nucleus (S. G. Parker, 2002; Selkirk, 1984). The sonority hierarchy is often represented as a numerical scale, e.g., 1 for lowest sonority, 9 for highest sonority (S. Parker, 2008). According to this scale, a syllable can be evaluated as following or violating the SSP by examining the sequence of sonority values assigned to each of its segments. For instance, the syllable /blif/ with sonority contour 2-6-**8**−3 would be considered well-formed under the SSP: sonority only increases from onset to nucleus, and only decreases toward the coda. On the other hand, the syllable /lbif/ (6-2-**8**−3) would be considered ill-formed under the SSP: at onset, sonority first dips from the first to the second phoneme, then increases toward the nucleus of the syllable.

Evidence from experimental studies suggests that the SSP plays an important role in speech processing, both in oral (Berent et al., 2007, 2011; Berent & Lennertz, 2010; Chen et al., 2024; Miozzo & Buchwald, 2013; S. Parker, 2017) and sign languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). Evidence from young children and adults points to a

consistent preference toward SSP-following over SSP-violating syllables, even if unattested in their native language. For instance, in an auditory perception task, English-native adults show a preference for the onset consonant cluster /bn/ (follows the SSP) over /lf/ (violates the SSP), both of which are unattested in English (Berent et al., 2011; Berent & Lennertz, 2010). Few-day-old newborns show different brain haemodynamic responses when listening to SSP-following (e.g., "bran") and violating (e.g., "lbif") syllables suggesting that precursors of the SSP are already present at birth (Gomez et al., 2014). Such a preference for SSP-following sequences of phonemes, even in unattested materials or in the absence of extensive linguistic experience, has been considered as supporting evidence for the status of the SSP as a linguistic universal, which humans are equipped with from birth. From a developmental viewpoint, innate knowledge of sonority constraints would provide humans with an advantage to quickly parse syllables in the speech signal into processing units (i.e., syllables), an essential step to process continuous speech stream, especially at the earliest stages of language acquisition.

The universal status of the SSP is however still an ongoing debate. While many languages seem to comply with the SSP more often than not (e.g., Greenberg, 1978; Frisch, 2015; Kreitman, 2012), violations of this principle are common across many languages. Virtually all languages show at least one instance of SSP violations (Yin et al., 2023). A clear example is the presence in many languages of onset consonant clusters of the type *initial fricative-stop consonant* (e.g., /ʃt/, as in the German word "Stein") or its inverse counterpart in coda *stop consonant-fricative* (e.g., /ks/, as in the English word "clicks") (Kawasaki-Fukumori et al., 1997). Languages that allow for fairly more complex consonant clusters at syllabic onset tend so show violations of the SSP more often. This is the case of Russian, which allows for clusters up to three consonants at syllabic onset (e.g., "взгляд" /vzglyad/ [glance]) (Halle & Jones, 1971; Itô, 1982) or Georgian, in which onset consonant clusters like /vrtʰ/ are attested (e.g., "მწვრთნელი" /mt͡sʼvrtʰneli/ [trainer, coach]) (Crouch et al., 2023). In some other, previously

neglected languages, the proportion of SSP-violating consonant clusters may simply be higher than what had been estimated in languages traditionally over-represented in linguistic studies (e.g., Lelepa Sun & Billington, 2024; Modern Standard Arabic Al Tamimi & Al Shboul, 2013). In light of these findings, the extent to which SSP-following syllables are more frequent than SSP-violating syllables is unclear.

Previous studies have investigated the prevalence of the SSP by counting the number of syllables or consonant clusters that followed or violated the SSP, or simply establishing the presence or absence of the latter in the repertoire of a given language or collection of languages. While this approach provides information about a given language's permissiveness for violations of the SSP (or lack thereof), it does not provide a direct estimate of how often speakers of such languages encounter violations of the SSP in natural speech. This is because the prevalence of a specific phonological structure in the repertoire of a language does not map onto a one-to-one basis to its frequency in natural speech. Take the following imaginary example: a given language allows for 20 consonant clusters in total, one of which is the /st/ consonant cluster. While this consonant cluster represents a big portion of this language's phonotactical repertoire (5%), how frequently speakers of that language encounter the /st/ sequence depends on at least two additional factors: the number of word-forms in which /st/ is included, and the lexical frequency of each of those word-forms. If /st/ occurs in few words in the language's lexicon or occurs only in mostly low-frequency words, the actual prevalence of the /st/ consonant cluster in natural speech would be lower than its share in the languages' phonological repertoire suggests. An estimation of the prevalence of SSP-following structures in natural speech should therefore consider some measure of the frequency of SSP-following structures in natural speech, instead of exclusively reporting its existence or absence, or the proportion it represents in a language's phonological repertpoire.

To address this issue, we used a Bayesian modelling approach to estimate the prevalence of the SSP in a large dataset of 35,848 syllable types across 42, providing estimates of the proportion of SSP-following syllables in each language, adjusted for the relative frequency of each syllable in natural speech.

Methods

Raw data, reproducible code, and project documentation are available in the Open Science Framework (https://osf.io/yeqms) and GitHub (https://github.com/NeuroDevCo/ssp-corpus).

Data collection and preprocessing

We scraped all phonological transcriptions from word entries in the Wiktionary database. Wiktionary is an open-source, free database of lexical entries containing. Wiktionary contains 10,050,751 lexical entries from 8,244 languages contributed by 4,463,976 total users (as of June 10th, 2026). Some lexical entries in Wiktionary also include a phonological in IPA format or phonetic transcription of the word-form in a particular language. We used the `wikipron` Python package (version 1.3.3, Lee et al., 2020) to scrape all available phonological transcriptions from all 7,859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in ~1.7 million phonological transcriptions belonging to 900 languages. We filtered out languages with less than 100 phonological transcriptions available, resulting in a dataset with 731,145 word-forms from 192 languages.

We extracted available lexical frequencies for each word-form using the `wordfreq` Python package (Speer, 2022), which provides an interface to retrieve lexical frequency data for words in over 40 languages. These lexical frequencies were calculated from a variety of sources according to their availability for each language, including Wikipedia, subtitles (e.g., OPUS Open subtitles 2018 and SUBTLEX), news, books, web text, Twitter, and Reddit, among others. At least three of these sources contribute to the calculation of lexical frequencies in each of the

available languages. When lexical frequencies were not available for a particular language (e.g., the ISO 639-3 code does not match any available language in `wordfreq`), an equivalent language was used, if available. Equivalent languages were searched according to their distance to the original language according to CLDR v38.1, as implemented[1]. The rationale behind using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and therefore lexical frequencies are expected to be reasonably equivalent across such language pairs. Word-forms with missing lexical frequency scores were excluded from the dataset. The resulting dataset included 368,697 word-forms from 57 languages.

We then segmented word-form phonological transcriptions into syllable candidates where possible. If available, we used explicit syllable boundary markers ("`.`") or implicity syllable markers ("`ˈ`" for primary stress, "`ˌ`" for secondary stress). In the absence of syllabic markers, we assumed the word to be monosyllabic. We then classified the segments inside each syllable candidate into three categories: consonants (`C`), vowels (`V`), and semi-vowels (`v`). Consecutive sequencies of vowels and semi-vowels (e.g., `V`, `VV`, `Vv`, `VVv`) where classified as vocalic clusters. Syllable candidates with more than one vocalic cluster were flagged as multisyllabic and excluded from the dataset. We also remove duplicated syllables within each language (e.g., if the syllable /ka/ appeared in Spanish words /ˈka.sa/ and /kaˈpo/ in Spanish, which would appear more than once in the Spanish dataset). The final result was a syllabic repertoire for each language. This dataset included 52,588 syllables from 42 languages from 8 distinct language families. To evaluate if each extracted syllable followed or violated the SSP, we assigned a sonority score to each segment in each syllable in a scale from 1 to 9, according to the `panphon` Python package

[1]The CLDR system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål → Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans → Dutch, Wu Chinese → Mandarin Chinese), etc.

(version 0.22.2, Mortensen et al., 2016). Syllables with more than one peak (i.e., local minimum) according to the sonority sequence were classified as SSP-violating (and as SSP-following otherwise). Syllables with a flat sonority contour (e.g., English words /ˈsfɪŋks/, [**3**, **3**, 8, 5, 1, 3]) were classified as SSP-following. We also run the same analysis after classifying flat sonority contours as SSP violating, and found equivalent results (see Supplementary Materials).

Table 1

Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically. For example, the syllable /ʻskul/ in Assamese shows a violation at onset, as sonority drops from /s/ (sonority index = 3) to /k/ (sonority index = 1) before rising again at nucleus /u/ (sonority index = 8).

Language	Syllable	Original word	Sonority index	
SSP-following				
Faroese	/aɻ/	altar /ˈaˌl̥.taɹ/	9-6	
Finnish	/berk/	Berg /ˈberk/	2-9-6-1	
Kankanaey	/bal/	balla /balˈla/	2-9-6	
Moksha	/alʲ/	альняка /ˈalʲ.nʲa.kə/	9-6	
Sanskrit	/be:/	कुबेर /ku.be:.ɾɐ/	2-9	
SSP-violating				
Assamese	/skul/	স্কুল /skul/	3-1-8-6	
German	/amts/	Amtsträger /ˈamtsˌtʁe:gɐ/	9-5-1-3	
English	/bi:dz/	beads /bi:dz/	2-8-2-4	
Serbian-Croatian-Bosnian	/bf̂:zo/	brzo /bf̂:zo/	2-6-4-9	
Mon	/hma/	ဘ္ဟာ /hma/	6-5-9	

To calculate syllabic frequencies, we aggregated the lexical frequency (frequency per million words) of each word each syllable was part of in the dataset (see Duchon et al. (2013) for a similar approach). The rationale behind this step is that each time a word is uttered, each of its syllables is also uttered. Therefore, by calculating the lexical frequency of each word a given syllable is involved in, an estimation of syllabic frequency can be derived. The final dataset comprised 52,588 syllables from 42 languages from 8 distinct language families.

Data analysis

We used a multilevel, Bayesian logistic regression model to estimate the probability of each syllable in the dataset following (or violating) the SSP, taking into account the language it belonged to, and its syllabic frequency. We modelled the response variable y_i (`1`, if the syllable was SSP-following, `0` otherwise) using a Bernoulli probabilistic distribution with probability parameter θ_i. We specified a linear regression model for θ_i, which included fixed regression coefficients for the intercept (α) and the slope of syllabic frequency (β), and random effects u_j and w_j for each fixed coefficient, respectively. We specified weakly informative priors for all parameters in the model (see Equation 1). To estimate the posterior distribution of the model used Hamiltonian Monte Carlo as implemented in PyMC3 (version 6.2.0, Patil et al., 2010), with eight chains of 2,000 iterations, 1,000 of which were warm-up iterations.

Likelihood:

Global prevalence of the SSP:

$$
\begin{aligned}
y_{ij} &\sim \text{Bernoulli}\ (\frac{\theta_{ij}}{1-\theta_{ij}}) \\
\theta_{ij} &= \alpha + u_j + (\beta + w_j) \cdot \text{Frequency}_{ij} \\
\\
\textbf{Prior}: \\
\alpha, \beta &\sim \text{Normal}\ (0, 1) \\
u_j, w_j &\sim \text{Normal}\ (0, \sigma_{u,w}) \\
\sigma_{u,w} &\sim \text{HalfNormal}\ (0.5)
\end{aligned}
\tag{1}
$$

Where:

i indexes each observation and j indexes the language it belongs to

y_{ij} indexes whether the syllable follows (1) or violates (0) the SSP

u_j, w_j are language-specific adjustments to α and β

Results

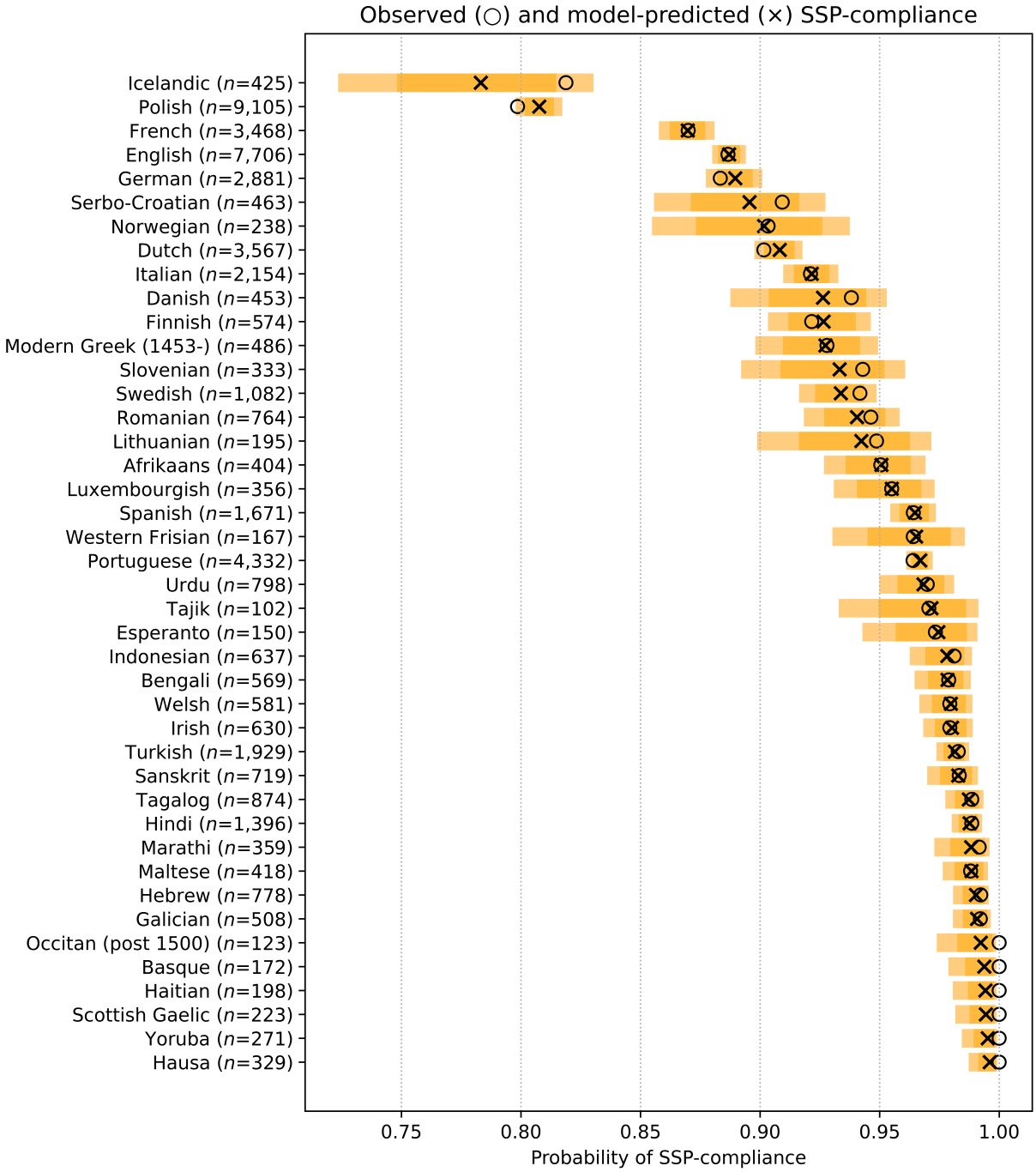
We report the median and 95% credible interval (CrI) of the posterior distribution of the model parameters of interest. The 95% CrI contains the 95% of the posterior distribution of the parameter. This interval can be interpreted as being 95% likely to contain the true value of the parameter, given the observed data and the prior distribution of the model (Kruschke & Liddell, 2018).

The global proportion of SSP-following syllables across languages was θ = 0.966 (95% CrI = [0.953, 0.977]), and the language-specific prevalence of the SSP ranged from 78.319% (Icelandic, 95% CrI = [72.353%, 81.730%]) to 99.605% (Hausa, 95% CrI = [98.720%, 99.918%]) (see Figure 1). Only 2.38% of the languages in the sample showed less than 80% of

SSP prevalence, 14.29% languages showed less than 90% of SSP prevalence, and 38.10% languages showed less than 95% of SSP prevalence.

Figure 1

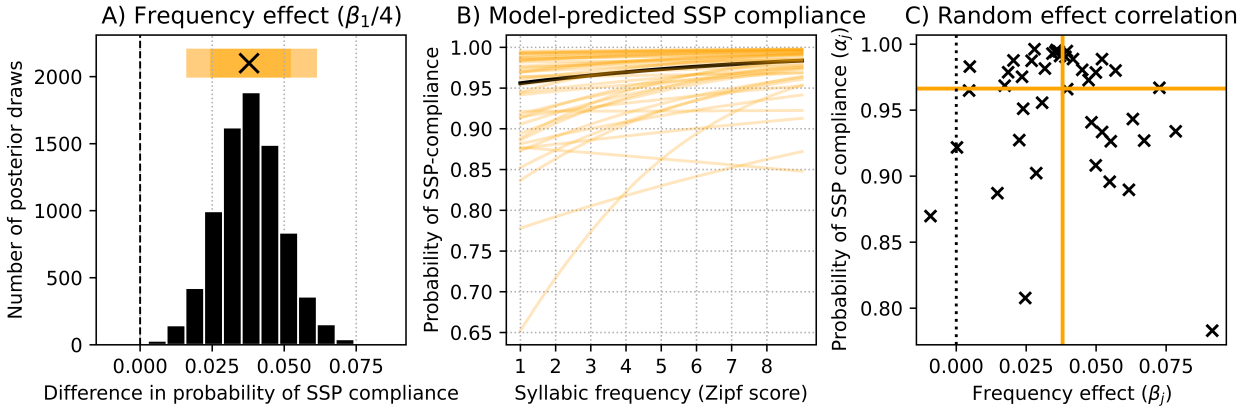
Observed (empty circles) and model-predicted (crosses) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% and 79% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language.



Syllabic frequency showed a positive, non-zero effect on the probability of SSP compliance (β =0.151, 95% CrI = [0.064, 0.246]), indicating that SSP-following syllables were, on average, more frequent than SSP-violating syllables. For every standard deviation increment in syllabic frequency (corresponding to 1.19 Zipf scores), the probability of SSP compliance increased a maximum of 3.782% (95% CrI = [1.602%, 6.139%]) (see Figure 2).

Figure 2

Summary of the model estimates. A) Posterior distribution of the fixed regression coefficient of the syllabic frequency predictor (β). Median and 95%, and 79% CrI of the distribution are indicated as a cross and intervals. B) Mean predictions of SSP compliance as a function of syllabif frequency. Predicted cross-language probability of SSP compliance (θ) is showed as a black line. Orange lines indicate language-specific predictions. C) Correlation between language-specific estimates of SSP compliance (u_j) and syllabic frequency effect (w_j). Each black cross indicates one language's estimates. For reference, cross-language estimates are indicated for each coefficient as orange lines.



Discussion

The present study was aimed at quantifying the cross-linguistic prevalence of the sonority sequencing principle (SSP). We gathered gathered a large number of phonological transcriptions from the Wiktionary database, and derived a syllabic repertoire for each of the 42 languages from

8 language families. We evlauated the proportion of SSP-following syllables present in each syllabic repertoire, and used Bayesian modelling to estimate the cross-linguistic, and language-specific prevalence of the SSP. We then derived a measure of syllabic frequency for each syllable in the dataset, and compared the relative frequency with which SSP-following and SSP-violating syllables appeared in adult-directed speech (ADS) and child-directerd speech (CDS). Our results show that an overwhelming majority oof the syllables across languages follow the SSP, with very few languages showing less than 90% prevalence of the SSP in their syllabic repertoire. We also found that SSP-following syllables were more frequent than SSP-violating syllables in both ADS and CDS.

Syllables are a privileged linguistic unit in speech processing during the earliest stages of language acquisition (Bertoncini et al., 1988; Eimas, 1999; Jusczyk & Derrah, 1987). For instance, during as early as their first days of life humans are able to discriminate between word-forms based on the number of syllables, instead of relying on the number of phonemes or the duration of the produced utterance (Bijeljac-Babic et al., 1993). Newborns can also track the co-ocurrence of syllable-sized groups of speech sounds to chunk the continuous speech stream into word-like units (Fló et al., 2019, 2022), an ability that will later give them access to the precursors of a lexicon (Saffran et al., 1996; Saffran & Kirkham, 2018). Given the relatively limited experience newborns have accumulated with their native language, their ability to identify syllables in speech is remarkable.

Previous literature had pointed to the SSP as a possible universal constraint to syllabic structure. This was based on the numerous examples of languages whose syllabic repertoire systematically gravitated toward SSP-following structures. Evidence for a role of the SSP in the marking of syllabic structures has even been shown in Imdlawn Tashlhiyt dialect of Berber, in which syllabic nuclei may involve obstruent (voiceless) consonants (Dell & Elmedlaoui, 1985; Prince & Smolensky, 2004). Newborns and young infants are sensitive to certain constraints to

syllabic structure (e.g. Bertoncini & Mehler, 1981; Gomez et al., 2014; Moon & Fifer, 1990; Santolin et al., 2024).

the SSP cannot be taken as a hard linguistic universal, but rather as an overall preference toward certain linguistic structures that that virtually all languages display to a varying extent (Prince & Smolensky, 2004).

Under this account, the posibility of the SSP being universally available as a cue for syllabification at birth becomes a quantitative question: how frequently do syllables abide by the SSP in natural speech?

0) Recap of study motivation and aims. Highlight syllabic frequency measure (novely compared to previous literature).
1) Summary and interpretation of results: 1.1) Very high observed prevalence of SSP across languages. 1.2) Very few languages show less than 80% of SSP prevalence 1.3) Evidence that SSP-following syllables are slightly more frequent than SSP-violating syllables, both in CDS and ADS.
2) Comparison with previous estimates of SSP, with especulation about why results may differ: 2.1) Yin et al. (2023): they show that virtually all languages show violations of the SSP. We provide converging evidence, while emphasizing that, while such violations exist, they may play a marginal role in language acquisition. 2.2) Other studies showing lower prevalence of the SSP within or across languages: 2.2.1) Different data sources: phonological inventories of attested consonant clusters vs. syllabic repertoires, syllabic frequencies. 2.2.2) Possible bias in our dataset: discuss how we accounted for the dataset size bias, and how if differs from previous attempts to correct for it. 2.2.3) Al Tamimi and Al Shboul (2013): standard arabic shows low SSP following in codas.
3) Implications: 2.1) Evidence that the SSP may be available as a cue for syllabification (specially) during language acquisition 2.2) The present status does not argue in favour or

against the universality status of the SSP, but does point to the SSP as a universally available, functionally relevant phonological feature

4) Limitations: 4.1) No direct evidence of SSP prevalence in the context of language usage (ideally, we would transcribe and syllabify a corpus and simply calculate the proportion of SSP-following syllables). Currently not possible (few datasets are syllabified). 4.2) Althoug the present study includes a relatively large sample of languages from different language families, Indoeuropean (mostly Western) languages are overrepresented in the dataset, paralelling the same pattern observed in corpus analysis in Linguistics. 4.3) The classification of segments into vowels and consonants ignores that in some languages, certain consonantal phonemes can act like syllabic nuclei. This is the case of post-stress /n/ in the English word "cotton" /kɒt.n̩/.
5) Conclusions

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Appendix

Number of SSP-following and SSP-violating syllables per language and language family

Table A1

Number of SSP-following and SSP-violating syllables for each language in the dataset.

Language	Syllables	Words	SSP+	SSP-	%	Language	Syllables	Words	SSP+	SSP-	%
Haitian	198	848	198	0	100.0%	Spanish	1,671	106,961	1,611	60	96.4%
Basque	172	10,555	172	0	100.0%	Western Frisian	167	425	161	6	96.4%
Scottish Gaelic	223	1,112	223	0	100.0%	Portuguese	4,332	99,468	4,176	156	96.4%
Yoruba	271	1,311	271	0	100.0%	Moselle Franconian	356	860	340	16	95.5%
Hausa	329	738	329	0	100.0%	Afrikaans	404	619	384	20	95.0%
Occitan	123	243	123	0	100.0%	Lithuanian	195	279	185	10	94.9%
Modern Hebrew	778	948	772	6	99.2%	Romanian	764	2,126	723	41	94.6%
Galician	508	3,477	504	4	99.2%	Slovenian	333	1,704	314	19	94.3%
Marathi	359	703	356	3	99.2%	Swedish	1,082	1,624	1,019	63	94.2%
Tagalog	874	13,663	864	10	98.9%	Danish	453	1,399	425	28	93.8%
Hindi	1,396	5,072	1,380	16	98.9%	Modern Greek	486	3,104	451	35	92.8%
Maltese	418	3,777	413	5	98.8%	Finnish	574	73,343	529	45	92.2%
Sanskrit	719	2,831	707	12	98.3%	Italian	2,154	15,211	1,984	170	92.1%
Turkish	1,929	5,108	1,896	33	98.3%	Serbian-Croatian-Bosnian	463	8,822	421	42	90.9%
Standard Indonesian	637	3,851	625	12	98.1%	Norwegian	238	451	215	23	90.3%
Irish	630	4,682	617	13	97.9%	Dutch	3,567	11,340	3,216	351	90.2%
Welsh	581	5,150	569	12	97.9%	English	7,706	36,929	6,832	874	88.7%
Bengali	569	2,771	557	12	97.9%	German	2,881	13,142	2,545	336	88.3%
Esperanto	150	5,975	146	4	97.3%	French	3,468	17,659	3,017	451	87.0%
Tajik	102	258	99	3	97.1%	Icelandic	425	2,346	348	77	81.9%
Urdu	798	1,088	774	24	97.0%	Polish	9,105	123,564	7,271	1,834	79.9%